Session-VII: Crop Water Budget (CWB) Workshop: October 2nd week

Introduction to CWB workshop

Crop Water Budget (CWB) workshop is an important milestone in the Farmer Water Schools. The CWB exercise involves estimation of the groundwater balance based on the recharge and draft for the particular monsoon season. The estimation helps farmers make informed decisions on the crops to be sown.

FWS participants organize CWB workshops at the Hydrological Unit level. The workshop acts a platform for sharing information pertaining to water balance, cropping pattern, and groundwater dynamics of the HU. Following this, farmers discuss the crop plans for the ensuing season and make informed decisions in accordance with the available groundwater balance.

Program Schedule

S.No	Time	Content	Method	Materials	Expected outcomes
1	40 min.	Inauguration of exhibition stalls and sharing farmer's learning.	Gallery walk		Exhibition stalls are inaugurated by the chief guest and farmers share their learning from FWS.
2	10 min.	Objectives of Crop Water Budget workshop.	Role play	Role play script	Community/farmers see the relevance of CWB exercise.
3	25 min.	Basic information and area of HU.	Large group discussion	HU banner	Participants discuss the basic information of HU.
4	10 min.	Groundwater recharge (June to October).	Large group discussion	HU banner.	Participants discuss the amount of groundwater recharge through rainfall during Kharif season (June-October).
5	15 min.	Estimation of groundwater draft & balance for the end of Kharif Season.	Large group discussion	Banner.	Participants discuss the draft during Kharif season.
6	30 min.	Crop plans for Rabi & water requirement.	Large group discussion	Banner & crop details.	Participants discuss Rabi crop plans and water requirement.
7	25 min.	Projected recharge through rainfall & secondary water bodies.	Large group discussion	Banners.	Participants discuss groundwater recharge through projected rainfall and secondary sources of water bodies.
8	10 min.	Projected Groundwater Situation by end of May.	Large group discussion	Banners.	Participants discuss groundwater balance at the end of May.
9	15 min.	Decisions on crop plans by HUN & respective GMC members.	Large group discussion		Participants discuss about the available groundwater and take decisions on crop-plans.
10	5 min.	Vote of thanks.			

Learning objectives

By the end of the exercise, participants will be able to:

- Visit stalls,
- State the objectives of CWB exercise,
- Discuss the amount of recharge through rainfall for Kharif season i.e. June to October,
- Discuss groundwater draft during Kharif season and estimate the groundwater balance,
- Discuss crop plans for Rabi season i.e. November to May and estimate total water requirement for Rabi crops,
- Discuss the amount of recharge through projected rainfall for Rabi season,
- Discuss the amount of recharge through secondary water bodies, and
- Make decisions on Rabi crops based on the estimated groundwater balance.

Process

1. Inauguration of exhibition stalls and sharing farmer's learning Time: 40 min.

- After welcoming the chief guest to CWB workshop, request him/ her to inaugurate exhibition stalls.
- In each stall, farmer participants of FWS sessions share their experience and learning to the chief guest and other invitees.
- After the completion of gallery walk, invite all the participants to occupy the seats arranged at the venue.

2. Objectives of CWB exercise Time: 10 min.

Objective : Participants share the objectives of CWB exercise.

Content : Objectives of CWB exercise.

Method : Role-play

Materials : Role-play script.

Process

- Select two volunteers and ask them to prepare for the role. Give the script of the role play.
- Encourage them to perform the role play.

Role-play

(Ramanna is carrying the plough to the field. Venkanna notices this.)

Venkanna: Where are you going?

Ramanna : I am going to sow the seeds.

Venkanna: No doubt, sowing is important. But Sai Ganesh Hydrological Unit Network committee has organized a meeting for the farmers. We should attend the meeting.

Ramanna : What is the use of attending the meeting? Will they give us any information on seeds?

Venkanna: They will give us more valuable information than seeds, i.e. on water. Let us go and listen to our colleagues who are the participants of FWS.

Ramanna : More value than money! What is that?

Venkanna: They will tell us the amount of groundwater recharge, the water spent during Kharif season and the balance of water available in the ground for Rabi season.

Ramanna : Looks like it is useful to farmers. If we know the groundwater balance, we can plan for Rabi crops. Isn't it? Have you informed other farmers? Venkanna: Everyone has been informed. Most of them should be at the venue, by now. (Ramanna hands-over the plough to the nearby house owner and they both go to the meeting.)

- Ask participants what they observed in the role play and share the objectives of CWB workshop.

3. Basic information and area of the Hydrological Unit Time: 25 min.

Objective : Participants discuss basic information of the HU.

Content :

- Boundaries of the HU.
- Streams, direction of flow.
- Number of OB wells and rain gauge stations.
- Villages in the HU.
- Area of the HU.

Method : Large group discussion

Materials : Hydrological Unit (HU) banner.

Process

Basic information of the HU

- Present the groundwater scenario of the HU using timeline and appeal to farmers for effective management of groundwater system.
- Use the HU banner to show the ridge portion and valley portion of the HU.
- Discuss about the physical features of the HU i.e. boundaries of the HU, major streams, direction of flow, habitations, major crops in the HU, types of soils, rock types, number of observation wells, rain gauge stations and any other land marks in the HU. (Banner should also show observation borewells, rain gauge stations, habitations, streams and rivulets).



Estimation of Hydrological Unit area

- Share the purpose of calculating the area of HU.
- Share that in order to know the amount of groundwater recharge through rainfall specific to the HU, it is critical to know the area of the HU.
- Next, share how the area of the HU was estimated during the sessions using graphs and grids.
- The grid on the graph represents a given area-250,000 square meters.
- Adding all the grids falling within HU boundary in the graph will give the total area of the HU.

4. Groundwater Recharge (June-October) Time: 10 min.

Objective : Participants discuss the amount of groundwater recharge through rainfall during Kharif season (June-October).

Content :

- Average rainfall received during Kharif season.
- Estimation of groundwater recharge (June-October).

Method : Large group discussion

Materials : Hydrological Unit (HU) banner.

Process

- Stress the need for collection of rainfall data from established rain gauge stations in the HU.
- Demonstrate ways of collecting the rainfall data and precautions to be taken in measuring the water collected in the collecting jar.
- Demonstrate the process of taking averages and share the average rainfall received for Kharif season i.e. June to October.
- Next, share the types of rock formations and its extent of the HU.
- Share the rate of infiltration in the HU. Use the banner to help farmers understand the rate of infiltration.
- Next, share the amount of groundwater recharge in terms of 'litres'. Equate the quantity of water into number of calibrated drums (100 litres) and equate these calibrated drums to the village level water tanks (one water tank = 50,000 litres)
- Use the following visuals to discuss the content.



5. Estimation of groundwater draft and balance for the end of Kharif season

Time: 15 min.

Objective : Participants discuss the draft during Kharif season.

Content :

- Data required for estimation of groundwater draft:
  - Number of functional borewells in the HU,
  - Number of pumping days,
  - Number of pumping hours in a day, and
  - Average drum discharge.

Method : Large group discussion

Materials : Banners.

Process

- Before sharing the amount of groundwater draft, share with the participants about the data required to estimate the groundwater draft for Kharif season.
- The data includes:
  - Number of functional borewells in the HU,
  - Number of pumping days,
  - Number of pumping hours in a day, and
  - Average discharge.

- Share the estimation process of groundwater draft for Kharif season.
- Use the following picture to discuss the content.



- Share that groundwater draft is a product of the number of functional wells, average number of functional days, average number of functional hours in a day, and the average drum discharge.
- Groundwater draft = no. of functional wells X average no. of functional days X average number of functional hours in a day X average drum discharge.
- Share the groundwater balance by the end of Kharif season or October.

6. Crop plans for Rabi and water requirement Time: 30 min.

Objective : Participants discuss Rabi crop plans and water requirement.

Content :

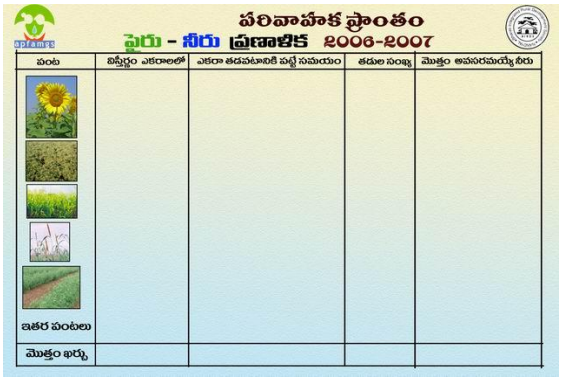
- Rabi crop plans and total crop area.
- Water requirement for Rabi crops.

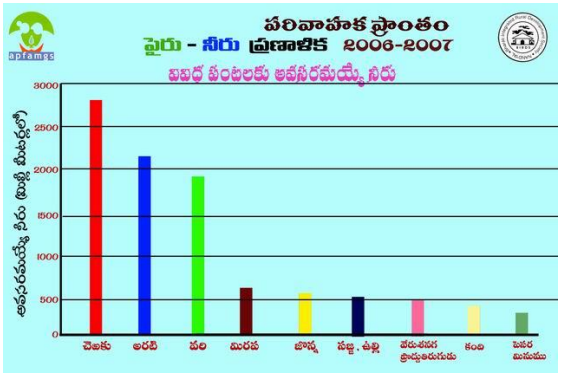
Method : Large group discussion.

Materials : Banners, and crop details.

Process

- Before starting the discussion, say these words: "We know that we are dependent on ground water to grow crops. Let us discuss about our Rabi crops and total water requirement to grow the planned crops".
- Use the following pictures to share the Rabi crops planned by the farmers in the HU.
- Share the process involved in collection of Rabi crop plans and the role played by the GMC and HUN in consolidation of village wise crop plans for the entire HU.





- Discuss the area under each crop and the water required for each crop.
- Discuss the total amount of water required for Rabi crops.

7. Projected Recharge through rainfall and secondary water bodies Time: 25 min.

Objective : Participants discuss groundwater recharge through projected rainfall and secondary sources of water bodies.

Content : Groundwater recharge through projected rainfall for November to May.
Groundwater recharge through secondary water bodies i.e. tanks, canals etc.

Method : Large group discussion

Materials : Banners.

Process

- So far, we have discussed the groundwater balance by the end of October and estimated the amount of water required for Rabi crops.
- Pose the following question to the large group if the groundwater balance is in deficit,
    - How do we meet the water demand? Are there any other sources for groundwater recharge?
- Wait for couple of minutes and listen to farmer responses.
- Next, share that we may receive some amount of rainfall and there is a chance of groundwater recharge.
- Discuss about the process of estimation of average projected rainfall for the period November to May.
- State that the projected rainfall is estimated based on the average rainfall received from November through May in the past 10 years. Share the amount of projected rainfall and the amount of groundwater recharge.
- Demonstrate the process of estimation of average rainfall.



- Ask participants to state sources of recharge through other means.
- Use the banner to encourage the discussion and share the amount of groundwater recharge through secondary water bodies.

8. Projected groundwater situation by the end of May Time: 10 min.

Objective : Participants discuss groundwater balance at the end of May.

Content : Groundwater balance at the end of May.

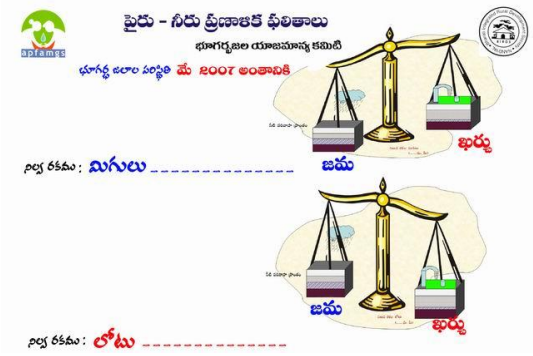
Method : Large group discussion

Materials : Banners.

Process

- Remind participants about the groundwater balance at the end of Kharif season and total water required for Rabi crops.
- If the amount of groundwater balance is negative, share the total groundwater recharge through projected rainfall and secondary water bodies, and
- Share the total water requirement for Rabi crops. Then, subtract the amount of water required from amount of recharge and share the projected groundwater balance by the end of May.





9. Decisions on crop-plans by HUN and respective GMC members Time: 15 min.

Objective : Participants discuss about the available groundwater and take decisions on crop-plans.

Content :

- Rabi crop-plans.
- Farmer decisions.

Process

- Request participants to think about the available groundwater and crops planned for Rabi season.
- Encourage them to share their ideas on change of crops or adoption of alternate irrigation techniques or reduce crop area or what ever they think.
- Request FWS participants to share their decisions on their crop-plans after knowing the deficit in water balance.
- Request GMCs to follow-up in immediate monthly meeting and encourage the farmers to take decisions on crop-plans to make use of available groundwater.

10. Vote of thanks Time: 5 min.

Thank participants for their presence in the workshop and thank each and every member who involved directly and indirectly in the conduct of CWB workshop.